

COMPASS_FE_20221220

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R Markdown

#notes I don't have a system yet to compare old slopes to current slopes for that day #I don't have a system for eliminating wells if the sd is high - if you think one of your sample wells was off you would need to remove it manually...

#Files Required: #csv of standards as list not plate with columns: Fe(II), Abs, Curve, FE ; these are: Known std concentration, Absorbance, Curve type (High or Low), and FE Curve (FE_2 or FE_Total)
#csv of samples as list not plate with columns: Date, Plate, Well_ID, SampleID, Dil_Factor, Abs1, Abs2 ; these are Date of run, Plate ID, Well ID, Sample ID, Dilution factor, Absorbance 1 (Fe2 or first read), and Absorbance 2 (feTot or second read) # I save these files in a folder for each day of FE runs

Read in Standards and subset

```
stds <- read.csv("Raw Data/FE_20230110_Std.csv")
head(stds)
```

##	Date	Plate	Well.Number	ID	dilution.factor	Abs1	Abs2
## 1	20230110	Standards1	1	Fe(II) A		1 0.235	0.345
## 2	20230110	Standards1	2	Fe(II) A		1 0.195	0.332
## 3	20230110	Standards1	3	Fe(II) A		1 0.165	0.295
## 4	20230110	Standards1	4	Fe(II) 100		1 0.275	0.461
## 5	20230110	Standards1	5	Fe(II) 100		1 0.243	0.403
## 6	20230110	Standards1	6	Fe(II) 100		1 0.308	0.474

```
stdid <- read.csv("Raw Data/Std_IDs_20230110.csv")
head(stdid)
```

##	ID	Concentration	Fe_2_Low	Fe_2_High	Fe_3
## 1	0.5 HCL	0	Yes	Yes	Yes
## 2	Fe(II) A	60	Yes	Yes	No
## 3	Fe(II) B	45	Yes	Yes	No
## 4	Fe(II) C	30	Yes	Yes	No
## 5	Fe(II) D	18	Yes	Yes	No
## 6	Fe(II) E	9	Yes	Yes	No

```
stds1 <- merge(stdid, stds)
head(stds1)
```

##	ID	Concentration	Fe_2_Low	Fe_2_High	Fe_3	Date	Plate	Well.Number	
## 1	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	88
## 2	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	89
## 3	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	90
## 4	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	88
## 5	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	89
## 6	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	88
##	dilution.factor	Abs1	Abs2						
## 1	1	0.038	0.039						
## 2	1	0.039	0.039						
## 3	1	0.037	0.038						
## 4	1	0.038	0.038						
## 5	1	0.038	0.038						
## 6	1	0.038	0.038						

#subset by FE_2 and FE_Total

```
FE_2_stds <- stds1[stds1$Fe_2_High == "Yes", ]
head(FE_2_stds)
```

One factor level

##	ID	Concentration	Fe_2_Low	Fe_2_High	Fe_3	Date	Plate	Well.Number	
## 1	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	88
## 2	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	89
## 3	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	90
## 4	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	88
## 5	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	89
## 6	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	88
##	dilution.factor	Abs1	Abs2						
## 1	1	0.038	0.039						
## 2	1	0.039	0.039						
## 3	1	0.037	0.038						
## 4	1	0.038	0.038						
## 5	1	0.038	0.038						
## 6	1	0.038	0.038						

```
FE_Tot_stds <- stds1[stds1$Fe_3 == "Yes", ]
head(FE_Tot_stds)
```

One factor level

##	ID	Concentration	Fe_2_Low	Fe_2_High	Fe_3	Date	Plate	Well.Number	
## 1	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	88
## 2	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	89
## 3	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	90
## 4	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards2	88
## 5	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards3	89
## 6	0.5	HCL	0	Yes	Yes	Yes	20230110	Standards1	88
##	dilution.factor	Abs1	Abs2						
## 1	1	0.038	0.039						
## 2	1	0.039	0.039						
## 3	1	0.037	0.038						
## 4	1	0.038	0.038						
## 5	1	0.038	0.038						
## 6	1	0.038	0.038						

```
#subset curves for high and low
```

```
FE_2_stds_high <- FE_2_stds[FE_2_stds$Fe_2_High == "Yes", ]  
head(FE_2_stds_high)
```

```
# One factor
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3      Date      Plate Well.Number  
## 1 0.5 HCL           0      Yes      Yes Yes 20230110 Standards3           88  
## 2 0.5 HCL           0      Yes      Yes Yes 20230110 Standards2           89  
## 3 0.5 HCL           0      Yes      Yes Yes 20230110 Standards1           90  
## 4 0.5 HCL           0      Yes      Yes Yes 20230110 Standards2           88  
## 5 0.5 HCL           0      Yes      Yes Yes 20230110 Standards3           89  
## 6 0.5 HCL           0      Yes      Yes Yes 20230110 Standards1           88  
##  dilution.factor Abs1 Abs2  
## 1              1 0.038 0.039  
## 2              1 0.039 0.039  
## 3              1 0.037 0.038  
## 4              1 0.038 0.038  
## 5              1 0.038 0.038  
## 6              1 0.038 0.038
```

```
FE_2_stds_low <- FE_2_stds[FE_2_stds$Fe_2_Low == "Yes", ]  
head(FE_2_stds_low)
```

```
# One factor l
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3      Date      Plate Well.Number  
## 1 0.5 HCL           0      Yes      Yes Yes 20230110 Standards3           88  
## 2 0.5 HCL           0      Yes      Yes Yes 20230110 Standards2           89  
## 3 0.5 HCL           0      Yes      Yes Yes 20230110 Standards1           90  
## 4 0.5 HCL           0      Yes      Yes Yes 20230110 Standards2           88  
## 5 0.5 HCL           0      Yes      Yes Yes 20230110 Standards3           89  
## 6 0.5 HCL           0      Yes      Yes Yes 20230110 Standards1           88  
##  dilution.factor Abs1 Abs2  
## 1              1 0.038 0.039  
## 2              1 0.039 0.039  
## 3              1 0.037 0.038  
## 4              1 0.038 0.038  
## 5              1 0.038 0.038  
## 6              1 0.038 0.038
```

```
#FE_Tot_stds_high <- FE_Tot_stds[FE_Tot_stds$Curve == "High", ]  
#head(FE_Tot_stds_high)
```

```
# One fa
```

```
#FE_Tot_stds_low <- FE_Tot_stds[FE_Tot_stds$Curve == "Low", ]  
#head(FE_Tot_stds_low)
```

```
# One fact
```

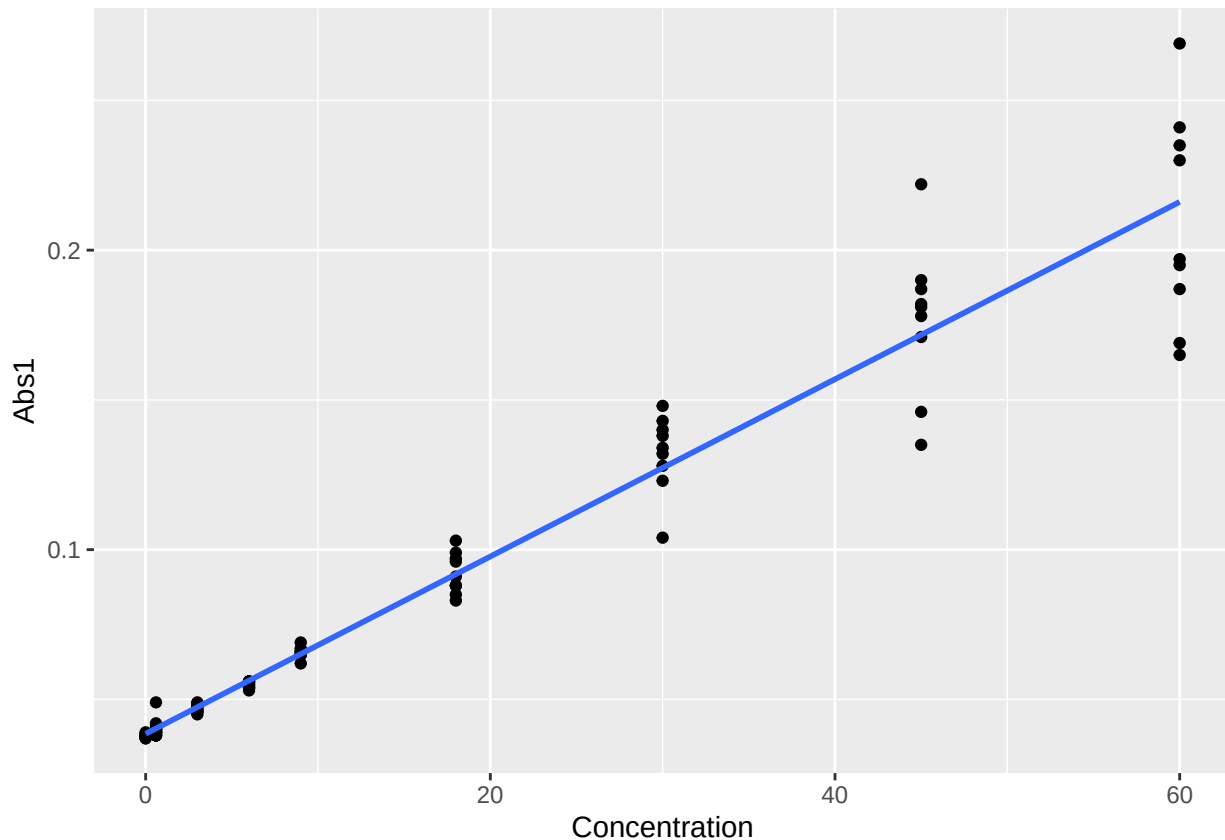
Plot standards

```
#Plot each of these and calculate the slope, intercept, and R2  
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
#Low curve for Fe(II)
FE_2_Low <- ggplot(FE_2_stds_low, aes(Concentration, Abs1)) + geom_point() + geom_smooth(method = "lm",
FE_2_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_Low_lm <- lm(FE_2_stds_low$Abs1 ~ FE_2_stds_low$Concentration)
summary(FE_2_Low_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_low$Abs1 ~ FE_2_stds_low$Concentration)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.051112 -0.001891 -0.000511  0.001229  0.052888
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.851e-02  1.750e-03   22.01  <2e-16 ***
## FE_2_stds_low$Concentration 2.960e-03  6.948e-05   42.60  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.01361 on 97 degrees of freedom
## Multiple R-squared:  0.9493, Adjusted R-squared:  0.9488
## F-statistic: 1815 on 1 and 97 DF,  p-value: < 2.2e-16
```

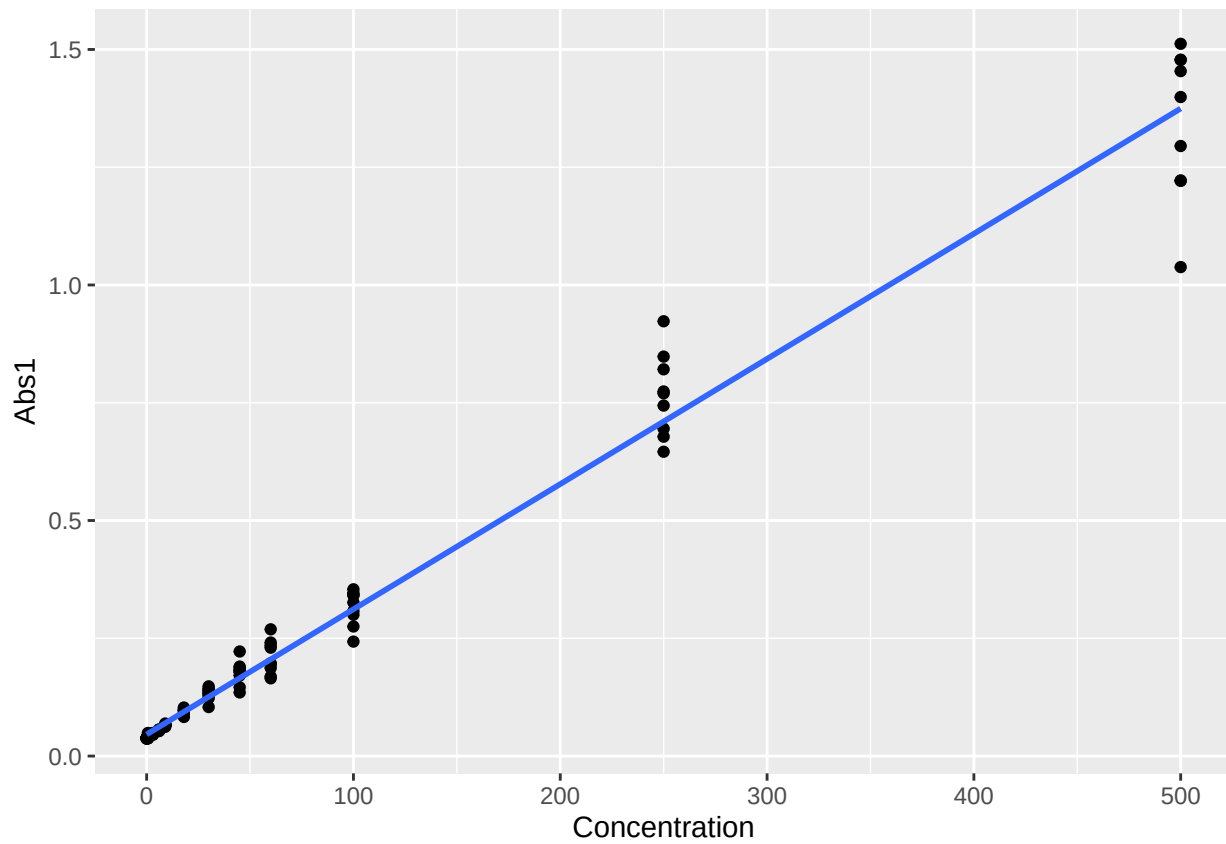
```
cf <- coef(FE_2_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes1 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes1$Curve <- "FE_2_Low"
Slopes1$R2 <- summary(FE_2_Low_lm)$adj.r.squared
Slopes1$Slope <- cf[2]
Slopes1$Intercept <- cf[1]
head(Slopes1)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_Low 0.9487503 0.002960016 0.03851094
```

```
#High curve for Fe(III)
FE_2_High <- ggplot(FE_2_stds_high, aes(Concentration, Abs1)) + geom_point() + geom_smooth(method = "lm")
FE_2_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_High_lm <- lm(FE_2_stds_high$Abs1 ~ FE_2_stds_high$Concentration)
summary(FE_2_High_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_high$Abs1 ~ FE_2_stds_high$Concentration)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33659 -0.00851 -0.00689  0.00786  0.21274
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.592e-02  5.287e-03   8.685 1.84e-14 ***
## FE_2_stds_high$Concentration 2.657e-03  3.446e-05  77.106 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05218 on 124 degrees of freedom
## Multiple R-squared:  0.9796, Adjusted R-squared:  0.9794
## F-statistic: 5945 on 1 and 124 DF, p-value: < 2.2e-16
```

```
cf <- coef(FE_2_High_lm)

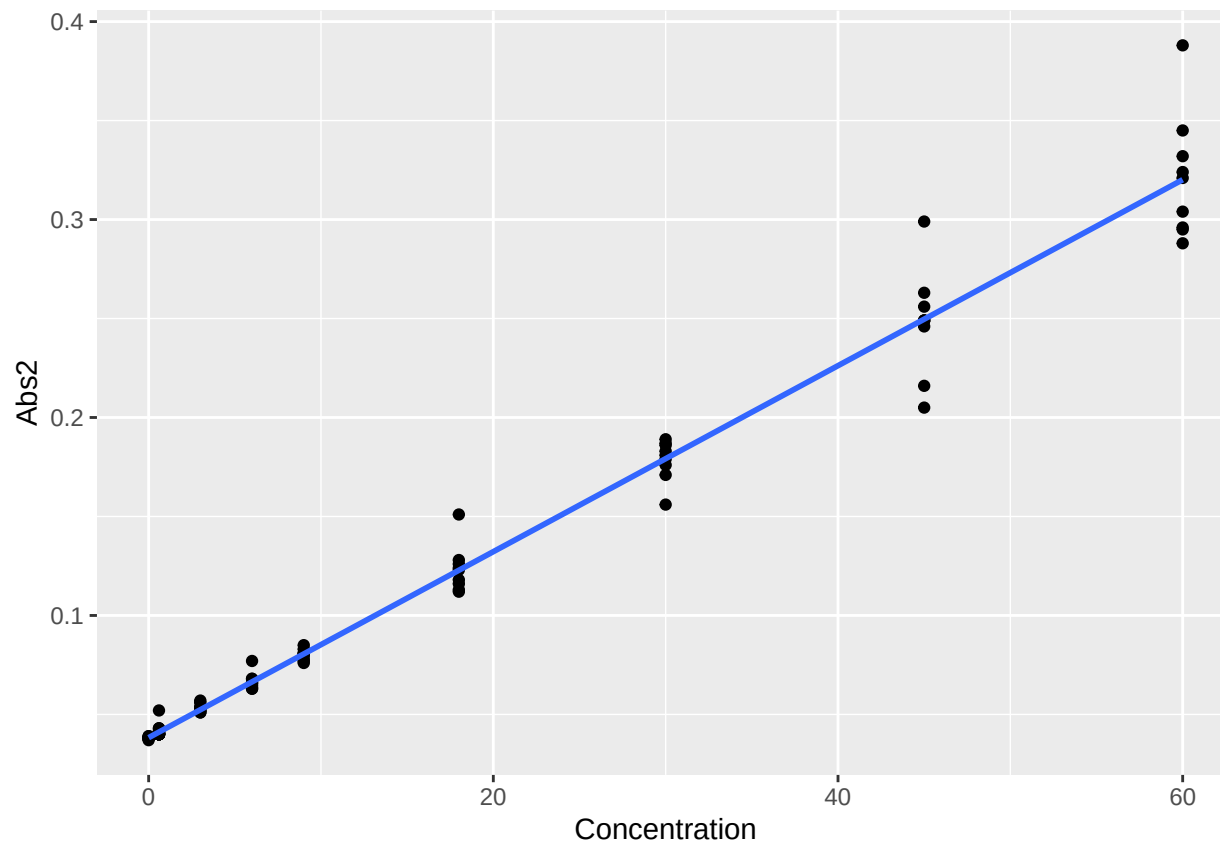
#create data frame with 1 rows and 0 columns
Slopes2 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes2$Curve <- "FE_2_High"
Slopes2$R2 <- summary(FE_2_High_lm)$adj.r.squared
Slopes2$Slope <- cf[2]
Slopes2$Intercept <- cf[1]
head(Slopes2)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_High 0.9794048 0.002657354 0.04591684
```

```
#Low curve for Fe(Total)
```

```
FE_Tot_Low <- ggplot(FE_2_stds_low, aes(Concentration, Abs2)) + geom_point() + geom_smooth(method = "lm")
FE_Tot_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_Low_lm <- lm(FE_2_stds_low$Abs2 ~ FE_2_stds_low$Concentration)
summary(FE_Tot_Low_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_low$Abs2 ~ FE_2_stds_low$Concentration)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.044658 -0.001084 -0.000266  0.001549  0.067878
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.827e-02  1.654e-03   23.14  <2e-16 ***
## FE_2_stds_low$Concentration 4.698e-03  6.566e-05   71.54  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01286 on 97 degrees of freedom
## Multiple R-squared:  0.9814, Adjusted R-squared:  0.9812
## F-statistic: 5118 on 1 and 97 DF,  p-value: < 2.2e-16
```

```
cf <- coef(FE_Tot_Low_lm)
```

```
#create data frame with 1 rows and 0 columns
```

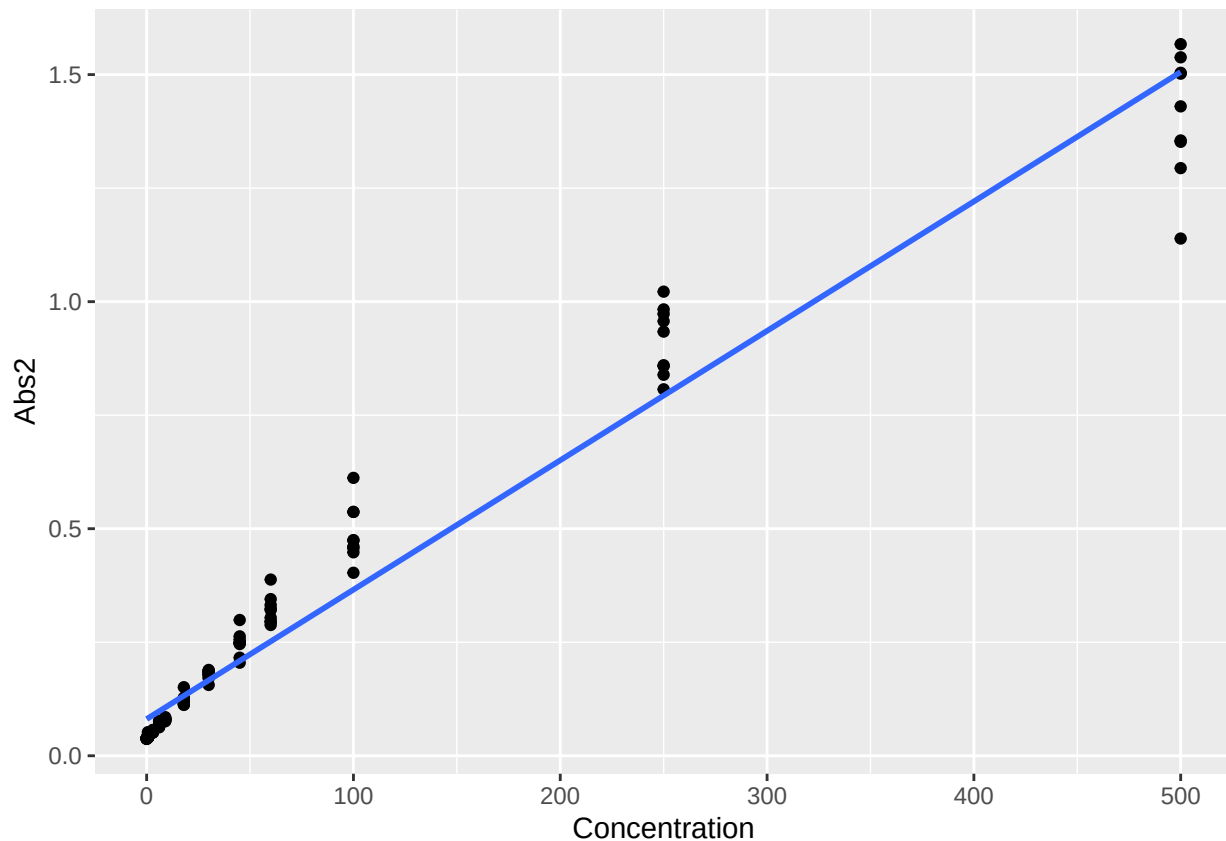
```
Slopes3 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes3$Curve <- "FE_Tot_Low"
Slopes3$R2 <- summary(FE_Tot_Low_lm)$adj.r.squared
Slopes3$Slope <- cf[2]
Slopes3$Intercept <- cf[1]
head(Slopes3)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_Tot_Low 0.9812076 0.004697602 0.03826555
```

```
#High curve for Fe(III)
```

```
FE_Tot_High <- ggplot(FE_2_stds_high, aes(Concentration, Abs2)) + geom_point() + geom_smooth(method = "lm")
FE_Tot_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_High_lm <- lm(FE_2_stds_high$Abs2 ~ FE_2_stds_high$Concentration)
summary(FE_Tot_High_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_high$Abs2 ~ FE_2_stds_high$Concentration)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.36659 -0.04151 -0.02595  0.03717  0.24624
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.080e-02  7.868e-03   10.27  <2e-16 ***
## FE_2_stds_high$Concentration 2.850e-03  5.129e-05   55.56  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07766 on 124 degrees of freedom
## Multiple R-squared:  0.9614, Adjusted R-squared:  0.9611
## F-statistic: 3087 on 1 and 124 DF, p-value: < 2.2e-16
```

```
cf <- coef(FE_Tot_High_lm)

#create data frame with 1 rows and 0 columns
Slopes4 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes4$Curve <- "FE_Tot_High"
Slopes4$R2 <- summary(FE_Tot_High_lm)$adj.r.squared
Slopes4$Slope <- cf[2]
Slopes4$Intercept <- cf[1]
head(Slopes4)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_Tot_High 0.9610705 0.002849573 0.08080246
```

```
#pull all the slopes together into one dataframe
Slopes <- rbind(Slopes1, Slopes2, Slopes3, Slopes4)
head(Slopes)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_Low 0.9487503 0.002960016 0.03851094
## 2 FE_2_High 0.9794048 0.002657354 0.04591684
## 3 FE_Tot_Low 0.9812076 0.004697602 0.03826555
## 4 FE_Tot_High 0.9610705 0.002849573 0.08080246
```

```
####
```

Read in Sample Data subset

```
dat <- read.csv("Raw Data/FE_20230110_Samples.csv")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230110 Samples7          1      Stds          1 1.256 1.377
## 2 20230110 Samples7          2      Stds          1 1.314 1.341
## 3 20230110 Samples7          3      Stds          1 1.222 1.213
## 4 20230110 Samples7          4      427          7 0.148 0.216
## 5 20230110 Samples7          5      427          7 0.138 0.211
## 6 20230110 Samples7          6      427          7 0.140 0.214
```

```
## remove any lines that say stds in SampleID
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
dat <- dat %>% filter(!SampleID == 'stds')
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7          1      Stds              1 1.256 1.377
## 2 20230110 Samples7          2      Stds              1 1.314 1.341
## 3 20230110 Samples7          3      Stds              1 1.222 1.213
## 4 20230110 Samples7          4      427              7 0.148 0.216
## 5 20230110 Samples7          5      427              7 0.138 0.211
## 6 20230110 Samples7          6      427              7 0.140 0.214
```

```
#Now flag any Abs1 that are above the 60uM standard absorbance (0.2)
dat$FE_2_Curve <- ifelse(dat$Abs1 >0.2, "High", "Low")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7          1      Stds              1 1.256 1.377
## 2 20230110 Samples7          2      Stds              1 1.314 1.341
## 3 20230110 Samples7          3      Stds              1 1.222 1.213
## 4 20230110 Samples7          4      427              7 0.148 0.216
## 5 20230110 Samples7          5      427              7 0.138 0.211
## 6 20230110 Samples7          6      427              7 0.140 0.214
##   FE_2_Curve
## 1      High
## 2      High
## 3      High
## 4      Low
## 5      Low
## 6      Low
```

```
#Now flag any Abs2 that are above the 60uM standard absorbance (0.3)
dat$FE_Tot_Curve <- ifelse(dat$Abs2 >0.3, "High", "Low")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7          1      Stds              1 1.256 1.377
```

```
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve
## 1      High      High
## 2      High      High
## 3      High      High
## 4      Low      Low
## 5      Low      Low
## 6      Low      Low
```

Subset Data and Calculate Concentrations

```
#Calculate concentrations of Fe(II)
dat$FE_2_Conc <- ifelse(dat$Abs1 > 0.2, (dat$Abs1 - Slopes2$Intercept) / Slopes2$Slope, (dat$Abs1 - Slopes1$Intercept) / Slopes1$Slope,
head(dat))
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230110 Samples7      1      Stds      1 1.256 1.377
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc
## 1      High      High 455.37139
## 2      High      High 477.19761
## 3      High      High 442.57671
## 4      Low      Low 36.98935
## 5      Low      Low 33.61099
## 6      Low      Low 34.28666
```

```
#calculate concentrations of Fe(total)
dat$FE_Tot_Conc <- ifelse(dat$Abs2 > 0.3, (dat$Abs2 - Slopes4$Intercept) / Slopes4$Slope,
head(dat))
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230110 Samples7      1      Stds      1 1.256 1.377
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc
## 1      High      High 455.37139 454.87430
## 2      High      High 477.19761 442.24083
## 3      High      High 442.57671 397.32182
## 4      Low      Low 36.98935 37.83515
## 5      Low      Low 33.61099 36.77077
## 6      Low      Low 34.28666 37.40940
```

```

#Account for the dilution factor
dat$FE_2_Conc_DilCorr <- dat$FE_2_Conc*dat$dilution.factor

dat$FE_Tot_Conc_DilCorr <- dat$FE_Tot_Conc*dat$dilution.factor

head(dat)

```

```

##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7         1      Stds             1 1.256 1.377
## 2 20230110 Samples7         2      Stds             1 1.314 1.341
## 3 20230110 Samples7         3      Stds             1 1.222 1.213
## 4 20230110 Samples7         4       427             7 0.148 0.216
## 5 20230110 Samples7         5       427             7 0.138 0.211
## 6 20230110 Samples7         6       427             7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      High      High 455.37139   454.87430      455.3714
## 2      High      High 477.19761   442.24083      477.1976
## 3      High      High 442.57671   397.32182      442.5767
## 4       Low       Low  36.98935    37.83515      258.9254
## 5       Low       Low  33.61099    36.77077      235.2769
## 6       Low       Low  34.28666    37.40940      240.0066
##   FE_Tot_Conc_DilCorr
## 1              454.8743
## 2              442.2408
## 3              397.3218
## 4              264.8460
## 5              257.3954
## 6              261.8658

```

Calculate Fe(III)

```

#Calculate concentrations of Fe(III)
dat$FE_3_Conc <- dat$FE_Tot_Conc_DilCorr-dat$FE_2_Conc_DilCorr

#Use ifelse to make any negative values equal to zero
dat$FE_2_Conc_Final <- ifelse(dat$FE_2_Conc_DilCorr <0, 0, dat$FE_2_Conc_DilCorr)

dat$FE_3_Conc_Final <- ifelse(dat$FE_3_Conc <0, 0, dat$FE_3_Conc)

#Use ifelse to make any negative values equal to zero
dat$FE_2_Info <- ifelse(dat$FE_2_Conc_DilCorr <0, 'bdl', 'within range')

#Use ifelse to make any negative values equal to zero
dat$FE_3_Info <- ifelse(dat$FE_3_Conc <0, 'bdl', 'within range')

head(dat)

```

```

##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7         1      Stds             1 1.256 1.377
## 2 20230110 Samples7         2      Stds             1 1.314 1.341
## 3 20230110 Samples7         3      Stds             1 1.222 1.213

```

```
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      High      High 455.37139   454.87430   455.3714
## 2      High      High 477.19761   442.24083   477.1976
## 3      High      High 442.57671   397.32182   442.5767
## 4      Low       Low  36.98935    37.83515    258.9254
## 5      Low       Low  33.61099    36.77077    235.2769
## 6      Low       Low  34.28666    37.40940    240.0066
##   FE_Tot_Conc_DilCorr   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final   FE_2_Info
## 1           454.8743  -0.4970927       455.3714      0.000000 within range
## 2           442.2408 -34.9567838       477.1976      0.000000 within range
## 3           397.3218 -45.2548941       442.5767      0.000000 within range
## 4           264.8460   5.9206015       258.9254      5.920601 within range
## 5           257.3954  22.1185101       235.2769     22.118510 within range
## 6           261.8658  21.8591724       240.0066     21.859172 within range
##       FE_3_Info
## 1         bdl
## 2         bdl
## 3         bdl
## 4 within range
## 5 within range
## 6 within range
```

Calculate Averages across wells and std. dev.

```
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7      1      Stds      1 1.256 1.377
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      High      High 455.37139   454.87430   455.3714
## 2      High      High 477.19761   442.24083   477.1976
## 3      High      High 442.57671   397.32182   442.5767
## 4      Low       Low  36.98935    37.83515    258.9254
## 5      Low       Low  33.61099    36.77077    235.2769
## 6      Low       Low  34.28666    37.40940    240.0066
##   FE_Tot_Conc_DilCorr   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final   FE_2_Info
## 1           454.8743  -0.4970927       455.3714      0.000000 within range
## 2           442.2408 -34.9567838       477.1976      0.000000 within range
## 3           397.3218 -45.2548941       442.5767      0.000000 within range
## 4           264.8460   5.9206015       258.9254      5.920601 within range
## 5           257.3954  22.1185101       235.2769     22.118510 within range
## 6           261.8658  21.8591724       240.0066     21.859172 within range
##       FE_3_Info
## 1         bdl
```

```
## 2          bdl
## 3          bdl
## 4 within range
## 5 within range
## 6 within range
```

```
#summarize by sampleID so that we can calculate the mean and std. dev. of the three wells
dat1 <- dat %>%
  group_by(SampleID) %>%
  summarise(FE_2_mean = mean(FE_2_Conc_Final), FE_2_sd = sd(FE_2_Conc_Final),
            FE_3_mean = mean(FE_3_Conc_Final), FE_3_sd = sd(FE_3_Conc_Final),
            FE_2_flag = first(FE_2_Info), FE_3_flag = first(FE_3_Info))

head(dat1)
```

```
## # A tibble: 6 x 7
##   SampleID FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag FE_3_flag
##   <chr>      <dbl>   <dbl>   <dbl>   <dbl> <chr>      <chr>
## 1 206        6025.   566.     0       0  within range bdl
## 2 212        7154.   513.    287.    32.6  within range within range
## 3 215        4584.   591.    532.    239.  within range within range
## 4 257        3346.   333.    987.    506.  within range within range
## 5 271         617.    48.7    43.0    41.2  within range within range
## 6 272         675.    73.8     0       0  within range bdl
```

Export full data then just final data

```
#Read in meta-data and sample IDs
IDs <- read.csv("Raw Data/FE_SampleIDs_2022.csv")
head(IDs)
```

```
##      Date Month Year Campaign Site Zone Depth Rep
## 1 20220502   May 2022   MonMon  MSM   WC    10   A
## 2 20220502   May 2022   MonMon  MSM   WC    10   B
## 3 20220502   May 2022   MonMon  MSM   WC    10   C
## 4 20220502   May 2022   MonMon  MSM   WC    20   A
## 5 20220502   May 2022   MonMon  MSM   WC    20   B
## 6 20220502   May 2022   MonMon  MSM   WC    20   C
##
##           Sample_Name SampleID
## 1 MSM_WC_PW_MonMon_20220502_SipA_10cm      654
## 2 MSM_WC_PW_MonMon_20220502_SipB_10cm      656
## 3 MSM_WC_PW_MonMon_20220502_SipC_10cm      658
## 4 MSM_WC_PW_MonMon_20220502_SipA_20cm      655
## 5 MSM_WC_PW_MonMon_20220502_SipB_20cm      657
## 6 MSM_WC_PW_MonMon_20220502_SipC_20cm      659
```

```
dat_IDs <- merge(IDs, dat1)
head(dat_IDs)
```

```
##   SampleID   Date   Month Year Campaign Site Zone Depth Rep
```

```
## 1      206 202210 October 2022   MonMon  MSM    UP    45    C
## 2      212 202210 October 2022   MonMon  MSM    TR    45    B
## 3      215 202210 October 2022   MonMon  MSM    TR    45    C
## 4      257 202210 October 2022   MonMon  MSM    TR     4    4
## 5      271 202210 October 2022   MonMon  MSM    WC     6    6
## 6      272 202210 October 2022   MonMon  MSM    WC     7    7
##          Sample_Name FE_2_mean   FE_2_sd FE_3_mean   FE_3_sd   FE_2_flag
## 1      MSM_UP_LysC_45cm 6025.2586 566.08703   0.00000    0.00000 within range
## 2      MSM_TR_LysB_45cm 7154.2010 513.33785 286.54985 32.56556 within range
## 3      MSM_TR_LysC_45cm 4583.9755 591.26842 531.86565 239.33238 within range
## 4 MSM_202210_TR_PEEPER_4 3345.9020 333.29304 987.36575 506.30610 within range
## 5 MSM_202210_WC_PEEPER_6 617.3398 48.69411 43.00152 41.24836 within range
## 6 MSM_202210_WC_PEEPER_7 674.7720 73.84248 0.00000 0.00000 within range
##          FE_3_flag
## 1              bdl
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6              bdl
```

```
#Read out the summarized data
head(dat_IDs)
```

```
## SampleID Date Month Year Campaign Site Zone Depth Rep
## 1      206 202210 October 2022   MonMon  MSM    UP    45    C
## 2      212 202210 October 2022   MonMon  MSM    TR    45    B
## 3      215 202210 October 2022   MonMon  MSM    TR    45    C
## 4      257 202210 October 2022   MonMon  MSM    TR     4    4
## 5      271 202210 October 2022   MonMon  MSM    WC     6    6
## 6      272 202210 October 2022   MonMon  MSM    WC     7    7
##          Sample_Name FE_2_mean   FE_2_sd FE_3_mean   FE_3_sd   FE_2_flag
## 1      MSM_UP_LysC_45cm 6025.2586 566.08703   0.00000    0.00000 within range
## 2      MSM_TR_LysB_45cm 7154.2010 513.33785 286.54985 32.56556 within range
## 3      MSM_TR_LysC_45cm 4583.9755 591.26842 531.86565 239.33238 within range
## 4 MSM_202210_TR_PEEPER_4 3345.9020 333.29304 987.36575 506.30610 within range
## 5 MSM_202210_WC_PEEPER_6 617.3398 48.69411 43.00152 41.24836 within range
## 6 MSM_202210_WC_PEEPER_7 674.7720 73.84248 0.00000 0.00000 within range
##          FE_3_flag
## 1              bdl
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6              bdl
```

```
write.csv(dat_IDs, file="Processed Data/FE_20230110_Summary_Data.csv")
```

```
#Read out all the data in dat
head(dat)
```

```
## Date Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230110 Samples7          1      Stds          1 1.256 1.377
```

```
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      High      High 455.37139   454.87430      455.3714
## 2      High      High 477.19761   442.24083      477.1976
## 3      High      High 442.57671   397.32182      442.5767
## 4      Low      Low  36.98935    37.83515      258.9254
## 5      Low      Low  33.61099    36.77077      235.2769
## 6      Low      Low  34.28666    37.40940      240.0066
##   FE_Tot_Conc_DilCorr   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final   FE_2_Info
## 1           454.8743  -0.4970927           455.3714      0.000000 within range
## 2           442.2408 -34.9567838           477.1976      0.000000 within range
## 3           397.3218 -45.2548941           442.5767      0.000000 within range
## 4           264.8460   5.9206015           258.9254      5.920601 within range
## 5           257.3954  22.1185101           235.2769     22.118510 within range
## 6           261.8658  21.8591724           240.0066     21.859172 within range
##       FE_3_Info
## 1         bdl
## 2         bdl
## 3         bdl
## 4 within range
## 5 within range
## 6 within range
```

```
write.csv(dat, file="Processed Data/FE_20230110_Data.csv")
```

```
#Now take out the absorbance and stuff
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230110 Samples7      1      Stds      1 1.256 1.377
## 2 20230110 Samples7      2      Stds      1 1.314 1.341
## 3 20230110 Samples7      3      Stds      1 1.222 1.213
## 4 20230110 Samples7      4      427      7 0.148 0.216
## 5 20230110 Samples7      5      427      7 0.138 0.211
## 6 20230110 Samples7      6      427      7 0.140 0.214
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      High      High 455.37139   454.87430      455.3714
## 2      High      High 477.19761   442.24083      477.1976
## 3      High      High 442.57671   397.32182      442.5767
## 4      Low      Low  36.98935    37.83515      258.9254
## 5      Low      Low  33.61099    36.77077      235.2769
## 6      Low      Low  34.28666    37.40940      240.0066
##   FE_Tot_Conc_DilCorr   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final   FE_2_Info
## 1           454.8743  -0.4970927           455.3714      0.000000 within range
## 2           442.2408 -34.9567838           477.1976      0.000000 within range
## 3           397.3218 -45.2548941           442.5767      0.000000 within range
## 4           264.8460   5.9206015           258.9254      5.920601 within range
## 5           257.3954  22.1185101           235.2769     22.118510 within range
## 6           261.8658  21.8591724           240.0066     21.859172 within range
```

```
##      FE_3_Info
## 1      bdl
## 2      bdl
## 3      bdl
## 4 within range
## 5 within range
## 6 within range
```

```
dat2 <- dat[ ,-(2:3)]
dat2 <- dat2[ ,-(3:12)]
head(dat2)
```

```
##      Date SampleID  FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final  FE_2_Info
## 1 20230110      Stds  -0.4970927      455.3714      0.000000 within range
## 2 20230110      Stds -34.9567838      477.1976      0.000000 within range
## 3 20230110      Stds -45.2548941      442.5767      0.000000 within range
## 4 20230110      427   5.9206015      258.9254      5.920601 within range
## 5 20230110      427  22.1185101      235.2769     22.118510 within range
## 6 20230110      427  21.8591724      240.0066     21.859172 within range
##      FE_3_Info
## 1      bdl
## 2      bdl
## 3      bdl
## 4 within range
## 5 within range
## 6 within range
```

```
write.csv(dat2, file="Processed Data/FE_20230110_short_Data.csv")
```

END